

# COMPUTER ORGANIZATION AND ARCHITECTURE (21CSS201T)

Unit 3 : Session 5 : SLO 2 — Booth's Algorithm (Simulation)

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## 1. Terminology Check (Remembering)

- 1.  $M \rightarrow$  b. The register that holds the Multiplicand
- 2.  $Q \rightarrow$  c. The register that holds the Multiplier
- 3.  $Q-1 \rightarrow$  a. The bit to the right of the LSB of  $Q$ , initialized to 0
- 4.  $A \rightarrow$  d. The register that holds the partial product, initialized to 0

## 2. Explaining the Steps (Understanding)

- If the bits are 10: Subtract  $M$  from  $A$  ( $A = A - M$ ), then perform an arithmetic right shift.
- If the bits are 01: Add  $M$  to  $A$  ( $A = A + M$ ), then perform an arithmetic right shift.
- If the bits are 00 or 11: perform only the arithmetic right shift (no arithmetic operation).

## 3. Positive Multiplication Simulation (Applying): $5 \times 3$ , $M = 0101$ , $Q = 0011$

- Initial:  $A = 0000$ ,  $Q = 0011$ ,  $Q-1 = 0$
- Cycle 1:  $Q_0Q_{-1} = 10 \rightarrow A = A - M = 1011$ ; shift  $\rightarrow A = 1101$ ,  $Q = 1001$ ,  $Q-1 = 1$
- Cycle 2:  $Q_0Q_{-1} = 11 \rightarrow$  no operation; shift  $\rightarrow A = 1110$ ,  $Q = 1100$ ,  $Q-1 = 1$
- Cycle 3:  $Q_0Q_{-1} = 01 \rightarrow A = A + M = 0011$ ; shift  $\rightarrow A = 0001$ ,  $Q = 1110$ ,  $Q-1 = 0$
- Cycle 4:  $Q_0Q_{-1} = 00 \rightarrow$  no operation; shift  $\rightarrow A = 0000$ ,  $Q = 1111$ ,  $Q-1 = 0$
- Final Answer ( $A$ ,  $Q$ ):  $0000\ 1111 = +15$  ( $5 \times 3 = 15$  ✓)

## 4. Negative Multiplication Simulation (Applying): $4 \times -3$ , $M = 0100$ , $-M = 1100$ , $Q = 1101$

- Initial:  $A = 0000$ ,  $Q = 1101$ ,  $Q-1 = 0$
- Cycle 1:  $Q_0Q_{-1} = 10 \rightarrow A = A - M = 1100$ ; shift  $\rightarrow A = 1110$ ,  $Q = 0110$ ,  $Q-1 = 1$
- Cycle 2:  $Q_0Q_{-1} = 01 \rightarrow A = A + M = 0010$ ; shift  $\rightarrow A = 0001$ ,  $Q = 0011$ ,  $Q-1 = 0$
- Cycle 3:  $Q_0Q_{-1} = 10 \rightarrow A = A - M = 1101$ ; shift  $\rightarrow A = 1110$ ,  $Q = 1001$ ,  $Q-1 = 1$
- Cycle 4:  $Q_0Q_{-1} = 11 \rightarrow$  no operation; shift  $\rightarrow A = 1111$ ,  $Q = 0100$ ,  $Q-1 = 1$
- Final Answer ( $A$ ,  $Q$ ):  $1111\ 0100 = -12$  in 2's complement ( $4 \times -3 = -12$  ✓)

## 5. Find the Mistake (Analyzing): $2 \times -2$ , $M = 0010$ , $Q = 1110$

- Mistake: the bits  $Q_0 Q_{-1}$  are 10, so the correct action is  $A = A - M$ , not  $A = A + M$  as written in Cycle 1.
- Corrected Cycle 1 row:  $M = 0010$ ,  $A = 1110 \rightarrow$  after shift  $A = 1111$ ,  $Q = 0111$ ,  $Q-1 = 0$ ; Action:  $A = A - M$  ( $A = 1110$ ), then arithmetic right shift.